

CS496 Quantum Tutorials - Solved Answers

Qubits, Gates, Entanglement, Teleportation, Shor, QFT, and Grover

Source tutorials solved	1.1 Qubits & Gates; 1.2 Quantum Gates Examples; 1.3 Teleportations; Tutorial 1 Quantum Computing; Tutorial 2 Quantum Computing
Total items covered	53 question-style items, including explicit questions, tables, circuit tasks, and gate examples
Main exam focus	Dirac notation, alpha/beta amplitudes, Hadamard, CNOT, Bell states, measurement, teleportation correction, QFT/Shor, Grover
Corrections made	Fixed common typo: Hadamard on $ 1\rangle$ uses $ 1\rangle = [0,1]^T$ and gives $(0\rangle - 1\rangle)/\sqrt{2}$.
Grade note	No guide can guarantee 100%, but every quantum tutorial item found in the reuploaded zip is answered and explained here.

High-Yield Formula Sheet

Concept	Must-know answer
General qubit	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$, with $ \alpha ^2 + \beta ^2 = 1$
Measurement probabilities	$P(0)= \alpha ^2$, $P(1)= \beta ^2$
Bloch sphere	$ \psi\rangle = \cos(\theta/2) 0\rangle + e^{i\phi}\sin(\theta/2) 1\rangle$
Hadamard	$H 0\rangle = (0\rangle + 1\rangle)/\sqrt{2}$, $H 1\rangle = (0\rangle - 1\rangle)/\sqrt{2}$
Bell state example	$ \Phi\rangle = (00\rangle + 11\rangle)/\sqrt{2}$
CNOT rule	target_out = target XOR control
Teleportation correction	Bob applies X if $c_1=1$ and Z if $c_0=1$
Grover speedup	$O(\sqrt{N})$ instead of $O(N)$
Shor idea	Factoring $\rightarrow$ period finding of $f(x)=a^x \bmod N \rightarrow \gcd(a^{r/2} \pm 1, N)$

Tutorial 1.1 - Qubits & Gates

Q1. What are qubits, and how do they work?

A qubit is the basic unit of information in quantum computing. A classical bit is only 0 or 1, but a qubit can be in a quantum state made from both basis states $|0\rangle$ and $|1\rangle$ at the same time. The state is written as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are probability amplitudes.

Qubits work through four main ideas: superposition lets them hold a combination of $|0\rangle$ and $|1\rangle$; measurement collapses the qubit to one classical result; entanglement links qubits so the total state cannot be described independently; and interference lets quantum algorithms increase useful answers and cancel wrong answers.

Exam key: Qubit = quantum bit; state can be $\alpha|0\rangle + \beta|1\rangle$; measurement gives 0 or 1 probabilistically.

Q2. What is superposition? What are the benefits of superposition?

Superposition means a qubit can exist as a linear combination of basis states before measurement: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. It does not mean we directly read many answers at once. It means the computation can evolve many probability amplitudes together.

Benefits: it allows quantum circuits to represent many possible computational paths at once, gives algorithms a larger state space to work with, and enables interference so correct answers become more likely while wrong answers become less likely.

Exam key: Superposition is useful only when combined with interference and measurement; it is not magic parallel classical output.

Q3. Explain how the superposition states of a qubit are represented and manipulated using Dirac notation. Describe the notation and the formula used.

Dirac notation uses kets to represent quantum states. The basic one-qubit basis states are $|0\rangle$ and $|1\rangle$. A general qubit state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β are complex probability amplitudes. The probability of measuring 0 is $|\alpha|^2$ and the probability of measuring 1 is $|\beta|^2$. The normalization rule is $|\alpha|^2 + |\beta|^2 = 1$.

Quantum gates manipulate the state by multiplying the state vector by a unitary matrix. For example, applying the Hadamard gate H to $|0\rangle$ gives $(|0\rangle + |1\rangle)/\sqrt{2}$. Applying H to $|1\rangle$ gives $(|0\rangle - |1\rangle)/\sqrt{2}$.

Exam key: Know the formula, what α/β mean, and the normalization rule.

Q4. What is a quantum gate? Provide examples of quantum gates.

A quantum gate is a reversible operation applied to one or more qubits. Mathematically, it is represented by a unitary matrix, which preserves total probability. Gates change probability amplitudes and phases.

Examples: X flips $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$; H creates superposition; Z flips phase; S and T add phase rotations; R_X , R_Y , R_Z rotate a qubit on the Bloch sphere; $CNOT/CX$ flips a target qubit if the control is $|1\rangle$; CZ adds a conditional phase; $SWAP$ exchanges two qubits; $CCX/Toffoli$ flips a target if two controls are both $|1\rangle$.

Exam key: Gate = unitary reversible operation; common examples: X , H , Z , $CX/CNOT$, $SWAP$, CCX .

Q5. What is a quantum circuit? Provide examples of quantum circuits.

A quantum circuit is an ordered sequence of quantum gates, measurements, and sometimes classical control. It describes how qubits are initialized, transformed, entangled, and measured.

Examples:

1. Superposition circuit: start with $|0\rangle$, apply H , then measure. The result is about 50% 0 and 50% 1.
2. Entanglement circuit: start with $|00\rangle$, apply H to qubit 0, then $CNOT$ from qubit 0 to qubit 1. The output is a Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$.
3. Teleportation circuit: create a Bell pair, perform Alice's $CNOT + H +$ measurements, send two classical bits, then Bob applies X/Z corrections.

Exam key: Circuit = gates in time order applied to qubits.

Q6. Explain how superposition is implemented using quantum gates.

Superposition is usually created using the Hadamard gate. If the qubit starts in $|0\rangle$, applying H gives:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

If the qubit starts in $|1\rangle$, applying H gives:

$$H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

After measuring $H|0\rangle$, the qubit gives 0 about half the time and 1 about half the time.

Exam key: H on $|0\rangle$ creates equal superposition $|+\rangle$.

Q7. What is entanglement? How does it arise in quantum physical systems?

Entanglement is a quantum relationship where the combined state of two or more qubits cannot be separated into independent states for each qubit. Measuring one qubit gives information about the other, even if the qubits are far apart. This does not allow faster-than-light communication, because useful information still requires a classical channel.

In physical systems, entanglement can arise when particles interact, are produced by the same process, or share correlated properties such as spin or polarization. In circuits, we deliberately create entanglement using gates such as H followed by $CNOT$.

Exam key: Entangled qubits share one joint state; they are not independently describable.

Q8. Explain how entanglement is implemented using quantum gates.

The standard Bell-state entanglement circuit starts with $|00\rangle$. First apply H to qubit 0:

$$|00\rangle \rightarrow (|00\rangle + |10\rangle)/\sqrt{2}$$

Then apply $CNOT$ with qubit 0 as control and qubit 1 as target:

$$(|00\rangle + |10\rangle)/\sqrt{2} \rightarrow (|00\rangle + |11\rangle)/\sqrt{2}$$

The final state is the Bell state $|\Phi^+\rangle$. It is entangled because it cannot be written as a separate state for qubit 0 times a separate state for qubit 1.

Exam key: Bell circuit: H on control, then $CNOT$ control $\rightarrow$ target.

Q9. What is the impact of adding a CNOT gate after a Hadamard gate?

Adding $CNOT$ after H turns a simple one-qubit superposition into a two-qubit entangled state. H creates a superposition in the control qubit. $CNOT$ copies the computational-basis correlation into the target without copying an arbitrary unknown quantum state.

Starting from $|00\rangle$, H then $CNOT$ produces $(|00\rangle + |11\rangle)/\sqrt{2}$. If one qubit is measured as 0, the other will be 0. If one is measured as 1, the other will be 1.

Exam key: H alone creates superposition; $H + CNOT$ creates entanglement.

Q10. What is the implication of entanglement for quantum networking?

Entanglement is a foundation for quantum networking because distant nodes can share correlated quantum states. This enables quantum teleportation, quantum key distribution, quantum repeaters, distributed quantum computing, and detection of eavesdropping in secure communication.

The important limitation: entanglement alone cannot send usable information faster than light. Teleportation and networking still require classical communication.

Exam key: Entanglement supports teleportation, QKD, and quantum internet concepts.

Q11. Prepare a table of common quantum gates, including the purpose of each gate, how it works, sample Qiskit code, and a text circuit diagram.

The table below covers the common gates you are likely to be asked about. The exact list of available gates depends on the framework and hardware backend, but these are the important course-level gates.

Exam key: For the exam, know X, H, Z, phase gates, rotations, CNOT/CX, CZ, SWAP, and Toffoli/CCX.

Gate	Purpose	How it works	Qiskit	Text diagram
I	Identity	Does nothing; useful as placeholder	qc.id(0)	q: -- --
X	Bit flip / NOT	Maps $ 0\rangle$ to $ 1\rangle$, $ 1\rangle$ to $ 0\rangle$	qc.x(0)	q: --X--
Y	Bit + phase flip	Maps $ 0\rangle$ to $i 1\rangle$, $ 1\rangle$ to $-i 0\rangle$	qc.y(0)	q: --Y--
Z	Phase flip	Leaves $ 0\rangle$; maps $ 1\rangle$ to $- 1\rangle$	qc.z(0)	q: --Z--
H	Superposition	Maps $ 0\rangle$ to $ +\rangle$, $ 1\rangle$ to $ -\rangle$	qc.h(0)	q: --H--
S	Phase $\pi/2$	Adds i phase to $ 1\rangle$	qc.s(0)	q: --S--
T	Phase $\pi/4$	Adds $e^{i(\pi/4)}$ phase to $ 1\rangle$	qc.t(0)	q: --T--
P(theta)	General phase	Adds $e^{i(\theta)}$ phase to $ 1\rangle$	qc.p(theta,0)	q: --P--
RX(theta)	X rotation	Rotates around X axis on Bloch sphere	qc.rx(theta,0)	q: --Rx--
RY(theta)	Y rotation	Rotates around Y axis on Bloch sphere	qc.ry(theta,0)	q: --Ry--
RZ(theta)	Z rotation	Rotates phase around Z axis	qc.rz(theta,0)	q: --Rz--
SX	Square-root X	Half X rotation; applying twice equals X	qc.sx(0)	q: --SX--
CX/CNOT	Controlled bit flip	Flips target if control is $ 1\rangle$	qc.cx(0,1)	c: --0-- t: --X--
CY	Controlled Y	Applies Y to target if control is $ 1\rangle$	qc.cy(0,1)	c: --0-- t: --Y--
CZ	Controlled phase	Adds phase flip if both qubits are $ 1\rangle$	qc.cz(0,1)	q0: --0-- q1: --Z--
CH	Controlled H	Applies H to target if control is $ 1\rangle$	qc.ch(0,1)	c: --0-- t: --H--
SWAP	Exchange qubits	Swaps the states of two qubits	qc.swap(0,1)	q0: --x-- q1: --x--
CCX/Toffoli	Controlled-controlled X	Flips target if both controls are $ 1\rangle$	qc.ccx(0,1,2)	c1:--0-- c2:--0-- t: --X--
CSWAP/Fredkin	Controlled swap	Swaps two targets if control is $ 1\rangle$	qc.cswap(0,1,2)	c: --0-- a: --x-- b: --x--
Measure	Read classical bit	Collapses qubit to 0 or 1	qc.measure(0,0)	q: --M--> c

Q12. Perform experimentation by building arbitrary circuits to investigate the impact of gate sequences.

Important experimental patterns:

1. H then H cancels: $H(H|0\rangle) = |0\rangle$. Two Hadamards return the qubit to the computational basis.
2. X then H changes the sign of the superposition: $H|1\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.
3. H then Z changes $|+\rangle$ to $|-\rangle$ by flipping the phase of $|1\rangle$.
4. H on one qubit followed by CNOT creates Bell-state entanglement.
5. Measuring before CNOT destroys the superposition, so the later circuit behaves more classically.

Use repeated shots to see probabilities. If a circuit is deterministic, counts should be close to 100% one output. If it has equal superposition, counts should split about 50/50.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

sim = AerSimulator()

def run(qc, shots=1024):
    tqc = transpile(qc, sim)
    return sim.run(tqc, shots=shots).result().get_counts()

# Experiment 1: H then H returns  $|0\rangle$ 
qc1 = QuantumCircuit(1, 1)
qc1.h(0)
qc1.h(0)
qc1.measure(0, 0)
```

```
print('H then H:', run(qc1))

# Experiment 2: Bell state
qc2 = QuantumCircuit(2, 2)
qc2.h(0)
qc2.cx(0, 1)
qc2.measure([0, 1], [0, 1])
print('Bell state:', run(qc2))
```

Exam key: Gate order matters. Some gates cancel; some change phase; H + CNOT creates entanglement.

Tutorial 1.2 - Quantum Gates Examples

Q13. Tutorial 1.2 gate example: What is the initial state of a qubit and why do we need gates?

Most basic circuits start a qubit in $|0\rangle$, whose vector representation is $[1, 0]^T$. A qubit can also be in $|1\rangle$, represented by $[0, 1]^T$. To make a qubit useful, we apply quantum gates to transform this initial state into another state, such as a superposition or an entangled state with another qubit.

Exam key: Initial basis states: $|0\rangle = [1, 0]^T$ and $|1\rangle = [0, 1]^T$.

Q14. Tutorial 1.2 gate example: Why must quantum gates be reversible and preserve probability amplitudes?

Quantum evolution in a closed system is unitary. That means the operation must be reversible and must preserve the total probability of all possible measurement outcomes. If a gate did not preserve probability, the state would no longer obey $|\alpha|^2 + |\beta|^2 = 1$. If it were not reversible, it would not represent valid quantum evolution.

Exam key: Valid quantum gates are unitary: reversible and probability-preserving.

Q15. Tutorial 1.2 gate example: Explain the NOT/X gate on $|0\rangle$ and $|1\rangle$.

The quantum NOT gate is the X gate. It flips computational basis states:

```
X|0> = |1>
X|1> = |0>
```

Its matrix is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Multiplying this matrix by $|0\rangle = [1, 0]^T$ gives $[0, 1]^T$, which is $|1\rangle$. Multiplying it by $|1\rangle = [0, 1]^T$ gives $[1, 0]^T$, which is $|0\rangle$.

```
from qiskit import QuantumCircuit
```

```
qc = QuantumCircuit(1, 1)
qc.x(0)
qc.measure(0, 0)
print(qc.draw(output='text'))
```

Exam key: X gate = quantum bit flip, same behavior as classical NOT on basis states.

Tutorial 1.3 - Teleportation, Shor, QFT, and Grover

Q16. Why is it difficult to send the state of a qubit directly on a network?

It is difficult because an unknown qubit state is fragile. Interaction with the environment causes decoherence, measurement collapses the state, and the no-cloning theorem says an unknown arbitrary quantum state cannot be copied perfectly. So classical network behavior like read, copy, amplify, and resend does not work for an unknown qubit.

Exam key: Direct sending is hard because of decoherence, measurement collapse, and no-cloning.

Q17. What is the solution for this problem? Describe the solution in detail, showing the main steps.

The solution is quantum teleportation. It transfers an unknown quantum state using a pre-shared entangled Bell pair and two classical bits.

Main steps:

1. Alice has the unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
2. Alice and Bob share a Bell pair. Alice holds one qubit; Bob holds the other.
3. Alice applies CNOT from $|\psi\rangle$ to her Bell qubit, then applies H to $|\psi\rangle$.
4. Alice measures her two qubits, producing two classical bits.
5. Alice sends the two classical bits to Bob over a normal classical channel.

6. Bob applies corrections to his qubit: X if the second bit is 1, Z if the first bit is 1.
7. Bob's qubit becomes the original $|\psi\rangle$. Alice's original state is destroyed, so no-cloning is not violated.

Exam key: Teleportation = entanglement + Bell measurement + two classical bits + Bob X/Z correction.

Q18. What is quantum teleportation and how does it work?

Quantum teleportation is not physical transportation of matter. It is the transfer of an unknown quantum state from one qubit to another distant qubit. The original state disappears from Alice's side and appears on Bob's qubit after correction. It works because the Bell pair creates a shared entangled resource. Alice's measurement projects the system into one of four cases, and the two classical bits tell Bob which correction to apply.

Exam key: Teleportation transfers state, not the physical particle.

Q19. Describe the quantum teleportation process and provide the detailed circuit diagram.

Use three qubits: q_0 is the unknown state $|\psi\rangle$, q_1 is Alice's half of the Bell pair, and q_2 is Bob's half.

Circuit idea:

1. Prepare Bell pair: $H(q_1)$, $CX(q_1, q_2)$.
2. Alice operation: $CX(q_0, q_1)$, $H(q_0)$.
3. Alice measurement: measure $q_0 \rightarrow c_0$ and $q_1 \rightarrow c_1$.
4. Bob correction: if $c_1=1$ apply $X(q_2)$; if $c_0=1$ apply $Z(q_2)$.

Text circuit:

```

q0 |psi> ----o----H----M c0 -----
              |      |
q1 |0> --H---X---M c1 --|-----
              |      |
q2 |0> ---X-----X^c1--Z^c0-- |psi>

```

The X and Z on Bob's qubit are classically controlled by Alice's measurement results.

Exam key: Correction rule: Bob applies X from Alice second bit, Z from Alice first bit.

Q20. What is the major factor behind the success of the quantum teleportation process?

The major factor is entanglement. The pre-shared Bell pair gives Alice and Bob a joint quantum resource. Alice's Bell measurement plus Bob's correction uses this entanglement to reconstruct the state at Bob's side. Classical communication is also required, but without the Bell-pair entanglement, teleportation cannot work.

Exam key: Core success factor = pre-shared entanglement.

Q21. Describe Shor factorization algorithm and explain how it improves on classical factorization algorithms.

Shor's algorithm factors a large composite integer N by reducing factoring to a period-finding problem. It chooses a number a that is coprime to N and studies the periodic function $f(x) = a^x \bmod N$. A quantum circuit finds the period r efficiently. Then classical post-processing computes $\gcd(a^{r/2}-1, N)$ and $\gcd(a^{r/2}+1, N)$ to recover non-trivial factors when the conditions are valid.

The improvement is that Shor gives an exponential speedup over known classical factoring approaches in the ideal fault-tolerant model. Classical factoring becomes very hard for large N , which is why RSA relies on it. Shor's algorithm can factor in polynomial time in the number of bits of N on a sufficiently powerful quantum computer.

Exam key: Shor = quantum period finding $\rightarrow$ classical gcd $\rightarrow$ factors; threatens RSA-style factoring security.

Q22. What is the role of Fourier transform in improving the complexity of Shor Algorithm?

The Quantum Fourier Transform (QFT) extracts the hidden period from the amplitudes created by modular exponentiation. After the circuit evaluates $a^x \bmod N$ over a superposition of x values, the results contain periodic structure. The inverse QFT transforms that periodic pattern into measurement peaks related to the period r .

This improves complexity because the quantum circuit does not test periods one by one. The QFT concentrates probability around values from which the period can be recovered efficiently using classical continued fractions.

Exam key: QFT turns hidden periodicity into measurable peaks.

Q23. Explain how a developer can define his own gates in Qiskit.

There are two common ways.

Method 1: Build a small QuantumCircuit and convert it to a gate using `to_gate()`. This is useful when the gate is made from existing operations.

Method 2: Define a custom unitary matrix using UnitaryGate. This is useful when you already know the mathematical matrix for the operation.

After creating the custom gate, append it to a circuit using `qc.append(gate, qubits)`.

```
from qiskit import QuantumCircuit
from qiskit.circuit.library import UnitaryGate
import numpy as np

# Method 1: build from existing gates
sub = QuantumCircuit(1, name='my_hx')
sub.h(0)
sub.x(0)
my_gate = sub.to_gate()

qc = QuantumCircuit(1)
qc.append(my_gate, [0])
print(qc.draw(output='text'))

# Method 2: define by matrix
x_matrix = np.array([[0, 1], [1, 0]])
custom_x = UnitaryGate(x_matrix, label='custom_X')
qc2 = QuantumCircuit(1)
qc2.append(custom_x, [0])
```

Exam key: Custom Qiskit gate = `circuit.to_gate()` or `UnitaryGate(matrix)`.

Q24. Describe in detail the circuit for Shor Algorithm.

A course-level Shor circuit has two quantum registers plus measurement.

1. Input: choose N to factor and a value a coprime to N .
2. First register: initialized to $|0\dots 0\rangle$. Apply H gates to create a superposition over x values.
3. Second register: initialized for modular exponentiation, often represented as $|1\rangle$.
4. Controlled modular exponentiation: compute $|x\rangle|1\rangle \rightarrow |x\rangle|a^x \bmod N\rangle$.
5. Apply inverse QFT to the first register to reveal the period information.
6. Measure the first register.
7. Classical post-processing: use continued fractions to estimate r , then compute $\gcd(a^{r/2} \pm 1, N)$.

Exam key: Circuit blocks: H superposition $\rightarrow$ controlled modular exponentiation $\rightarrow$ inverse QFT $\rightarrow$ measurement $\rightarrow$ gcd.

Q25. Describe Grover's algorithm and explain how it works.

Grover's algorithm searches an unsorted space of N possibilities in $O(\sqrt{N})$ oracle calls instead of $O(N)$ classical search. It does not instantly inspect all answers; it amplifies the amplitude of the marked answer.

Steps:

1. Apply H gates to create equal superposition over all candidate states.
2. Apply the oracle, which flips the phase of the correct/marked state.
3. Apply the diffuser, also called inversion about the mean, which increases the marked state's amplitude.
4. Repeat oracle + diffuser about $\sqrt{N}$ times.
5. Measure. The marked item has high probability.

Exam key: Grover = oracle phase mark + diffusion amplitude amplification.

Q26. What is the major factor behind the success of Grover's algorithm?

The major factor is amplitude amplification through quantum interference. The oracle creates a phase difference for the target state. The diffuser uses interference so the target state's amplitude increases constructively while non-target amplitudes are reduced relatively.

Exam key: Grover succeeds because of constructive/destructive interference.

Q27. Describe the circuit of Grover's search algorithm, showing the roles of each gate.

For n qubits:

1. H gates on all search qubits create equal superposition.
2. The oracle U_f marks the solution by applying a phase flip to the target state.
3. H/X/multi-controlled-Z/X/H gates often implement the diffuser. Conceptually, it reflects amplitudes around their average.
4. Measurement returns the candidate with the largest amplified probability.

Text circuit:

$|0\rangle \rightarrow H \rightarrow \text{Oracle} \rightarrow \text{Diffuser} \rightarrow \dots \rightarrow \text{Measure}$

$|0\rangle \rightarrow H \rightarrow \text{Oracle} \rightarrow \text{Diffuser} \rightarrow \dots \rightarrow \text{Measure}$

Repeat Oracle + Diffuser about $\text{floor}(\pi/4 * \sqrt{N})$ times for one marked item.

Exam key: Roles: H = superposition; Oracle = mark answer; Diffuser = amplify; Measure = read result.

Tutorial 1 - Quantum Computing

Q28. What is the first discovery that led to the birth of quantum mechanics and quantum computing? What is the implication of this for quantum research?

The first major discovery was Max Planck's 1900 quantum hypothesis: energy is exchanged in discrete packets called quanta, not as a perfectly continuous quantity. This introduced quantization and started the development of quantum mechanics.

Implication: quantum research studies nature at small scales where energy, matter, and measurement behave differently from classical intuition. It led to principles such as quantization, uncertainty, coherence, superposition, interference, and entanglement. These principles later became the basis for quantum computing.

Exam key: Planck 1900 $\rightarrow$ quantization $\rightarrow$ birth of quantum mechanics.

Q29. Describe each quantum principle and the impact of each on quantum computing: superposition, coherence, interference, entanglement.

Superposition: a qubit can be in $\alpha|0\rangle + \beta|1\rangle$. Impact: gives quantum algorithms a large state space.

Coherence: stable phase relationships between components of the wavefunction. Impact: allows superposition and interference to survive long enough for computation. Loss of coherence is decoherence and causes errors.

Interference: probability amplitudes combine constructively or destructively. Impact: algorithms use it to boost correct outcomes and suppress wrong outcomes.

Entanglement: multi-qubit states can be linked so they cannot be separated into independent qubit states. Impact: enables Bell states, teleportation, quantum networking, and algorithmic speedups.

Exam key: Coherence keeps the quantum behavior; interference makes algorithms useful; entanglement links qubits.

Q30. Describe how the state of a qubit is represented using Dirac notation. What are the major components of the state? What do the coefficients represent?

A qubit is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The major components are the basis states $|0\rangle$ and $|1\rangle$, and the coefficients α and β . The coefficients are complex probability amplitudes. $|\alpha|^2$ is the probability of measuring 0, and $|\beta|^2$ is the probability of measuring 1. They must satisfy $|\alpha|^2 + |\beta|^2 = 1$. The relative phase between α and β also affects interference.

Exam key: Coefficients are amplitudes, not direct probabilities; square their magnitudes to get probabilities.

Q31. Describe the corresponding radian angles representation of a qubit state, showing and explaining the parameters used.

A pure one-qubit state can be represented on the Bloch sphere as:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

θ is the polar angle from the +Z axis. $\theta = 0$ gives $|0\rangle$; $\theta = \pi$ gives $|1\rangle$. ϕ is the azimuth angle around the Z axis and controls the relative phase between $|0\rangle$ and $|1\rangle$. The Bloch sphere gives a geometric way to visualize one-qubit states.

Exam key: Bloch form: θ controls measurement probabilities; ϕ controls relative phase.

Q32. Explain how a qubit is taken into superposition mode. What is the gate used? Use Qiskit notation and a circuit diagram.

A qubit starting in $|0\rangle$ is taken into equal superposition by applying the Hadamard gate H:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Qiskit notation is `qc.h(0)`.

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
```

```
qc = QuantumCircuit(1)
```

```
qc.h(0)
state = Statevector.from_instruction(qc)
print(state)
print(qc.draw(output='text'))

# Text circuit:
# q0: |0> --H-- |>
```

Exam key: Hadamard gate creates equal superposition from $|0\rangle$.

Q33. What are the two major superposition states produced by the Hadamard gate based on the initial states?

The two major states are:

$$H|0\rangle = |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

$$H|1\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

$|+\rangle$ lies on the $+X$ direction of the Bloch sphere. $|-\rangle$ lies on the $-X$ direction.

Exam key: From $|0\rangle$ get $|+\rangle$; from $|1\rangle$ get $|-\rangle$.

Q34. Describe the Hadamard gate showing its unitary matrix representation and explain how this unitary representation is used to compute the two major superposition states, and show the positions of these on a Bloch sphere.

The Hadamard matrix is:

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Using $|0\rangle = [1, 0]^T$:

$$H|0\rangle = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^T = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$$

Using $|1\rangle = [0, 1]^T$:

$$H|1\rangle = (1/\sqrt{2}) \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^T = (|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$$

Bloch sphere: $|0\rangle$ is $+Z$, $|1\rangle$ is $-Z$, $|+\rangle$ is $+X$, and $|-\rangle$ is $-X$.

Exam key: Be careful: $|1\rangle$ vector is $[0, 1]^T$, not $[1, 0]^T$.

Q35. What is the major purpose of the measurement gate in quantum circuits and what are the major limitations of measurement gates?

Measurement converts quantum information into classical information. It returns a classical bit 0 or 1 according to the state's probabilities. In circuit diagrams, measured qubits are connected to classical bits.

Limitations:

1. Measurement collapses the state into a basis state.
2. One measurement does not reveal the full original quantum state.
3. Results are probabilistic, so many shots are used to estimate probabilities.
4. Measurement can destroy superposition and entanglement for the measured system.

Exam key: Measurement reads classical output but destroys the original quantum state information.

Entanglement circuit outputs (H on control, then CNOT)

Initial control,target	After H on control	After CNOT	Bell state
$ 00\rangle$	$(00\rangle + 10\rangle)/\sqrt{2}$	$(00\rangle + 11\rangle)/\sqrt{2}$	Phi+
$ 01\rangle$	$(01\rangle + 11\rangle)/\sqrt{2}$	$(01\rangle + 10\rangle)/\sqrt{2}$	Psi+
$ 10\rangle$	$(00\rangle - 10\rangle)/\sqrt{2}$	$(00\rangle - 11\rangle)/\sqrt{2}$	Phi-
$ 11\rangle$	$(01\rangle - 11\rangle)/\sqrt{2}$	$(01\rangle - 10\rangle)/\sqrt{2}$	Psi-

Q36. What is entanglement, and how is it implemented?

Entanglement is a joint quantum state where qubits cannot be described independently. It is commonly implemented by starting with $|00\rangle$, applying H to the first qubit, then applying CNOT with the first qubit as control and the second as target. Output: $(|00\rangle + |11\rangle)/\sqrt{2}$, one of the Bell states.

Exam key: Implement entanglement with H + CNOT.

Q37. Describe the role of the CNOT gate in implementing entanglement.

CNOT is a two-qubit gate. It flips the target qubit if and only if the control qubit is $|1\rangle$. When the control is in superposition, CNOT correlates the target with the control. That correlation turns a product state into an entangled state.

Example: after H on the control, the state is $(|00\rangle + |10\rangle)/\sqrt{2}$. CNOT maps $|10\rangle$ to $|11\rangle$, so the final state is $(|00\rangle + |11\rangle)/\sqrt{2}$.

Exam key: CNOT converts superposition into correlation/entanglement.

Q38. Construct the table of the possible outputs of the entanglement circuit based on all combinations of the initial states of the control and target qubits.

The table below assumes the entanglement circuit applies H to the control qubit first, then CNOT(control, target).

Exam key: Initial basis states produce the four Bell states.

Q39. It is generally known that the states of multi-qubit systems are the product of individual qubit states, except for one case. Provide an illustration using unitary matrix multiplication to compute states of simple two-qubit systems, and indicate the exception.

For independent qubits, the combined state is the tensor product. Example:

$$\begin{aligned} |0\rangle \text{ tensor } |+\rangle &= [1,0]^T \text{ tensor } (1/\sqrt{2})[1,1]^T \\ &= (1/\sqrt{2})[1,1,0,0]^T \\ &= (|00\rangle + |01\rangle)/\sqrt{2} \end{aligned}$$

This is separable because it can be written as a product of one-qubit states.

Exception: entangled states. Example:

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

There are no one-qubit states $|a\rangle$ and $|b\rangle$ such that $|a\rangle \text{ tensor } |b\rangle = |\Phi^+\rangle$. Therefore it is not a product state.

Exam key: Product rule works for separable states; entangled states are the exception.

Q40. Explain how the output state of an entanglement circuit is computed using the unitary representations of the gates. Compute the state before applying CNOT and after using CNOT.

Start with $|00\rangle = [1,0,0,0]^T$. Apply H to the first qubit, which means using H tensor I:

$$(H \text{ tensor } I)|00\rangle = (1/\sqrt{2})(|00\rangle + |10\rangle)$$

This is still separable: $(|0\rangle + |1\rangle)/\sqrt{2} \text{ tensor } |0\rangle$.

Now apply CNOT. CNOT leaves $|00\rangle$ unchanged and maps $|10\rangle$ to $|11\rangle$:

$$\text{CNOT} [(|00\rangle + |10\rangle)/\sqrt{2}] = (|00\rangle + |11\rangle)/\sqrt{2}$$

The output is the Bell state $|\Phi^+\rangle$, which is entangled.

Exam key: Before CNOT: superposition product state. After CNOT: Bell entangled state.

Q41. Build and test a Qiskit entanglement application that measures the states of the circuit before and after applying the CNOT gate.

The test should compare two circuits. Before CNOT, H on q0 gives q0 random and q1 fixed. After CNOT, the measurement outcomes are correlated: only 00 and 11 should appear ideally. Due to Qiskit classical bit ordering, printed bit strings may appear reversed depending on register mapping, but the correlation is the important point.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

sim = AerSimulator()

def counts(qc, shots=1024):
    tqc = transpile(qc, sim)
    return sim.run(tqc, shots=shots).result().get_counts()

# Before CNOT
before = QuantumCircuit(2, 2)
before.h(0)
before.measure([0, 1], [0, 1])
print('Before CNOT')
print(before.draw(output='text'))
print(counts(before))

# After CNOT
_after = QuantumCircuit(2, 2)
_after.h(0)
_after.cx(0, 1)
_after.measure([0, 1], [0, 1])
print('After CNOT')
print(_after.draw(output='text'))
print(counts(_after))
```

Exam key: Before CNOT: independent/random q0. After CNOT: correlated 00/11 Bell measurement.

Tutorial 2 - Quantum Computing: Teleportation + Bell States

Q42. What are the vector representations of states $|0\rangle$ and $|1\rangle$?

The computational basis vectors are:

$$|0\rangle = [1, 0]^T$$

$$|1\rangle = [0, 1]^T$$

This means $|0\rangle$ has amplitude 1 for state 0 and amplitude 0 for state 1. $|1\rangle$ has amplitude 0 for state 0 and amplitude 1 for state 1.

Exam key: $|0\rangle$ = column vector (1,0); $|1\rangle$ = column vector (0,1).

Q43. The Hadamard gate puts a qubit into superposition by transforming $|0\rangle$ and $|1\rangle$. What expressions result from transforming $|0\rangle$ and $|1\rangle$?

Hadamard transformations:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$$

$$H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$$

Exam key: Know both: $|+\rangle$ has plus sign, $|-\rangle$ has minus sign.

CNOT truth table

Control in	Target in	Control out	Target out
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$

Q44. The expression resulting from adding the transformations of $|0\rangle$ and $|1\rangle$ above describes the qubit superposition state. What is alpha and what is beta in this expression?

The general qubit superposition state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

alpha is the amplitude of $|0\rangle$, and beta is the amplitude of $|1\rangle$. Their squared magnitudes are probabilities: $P(0)=|\alpha|^2$ and $P(1)=|\beta|^2$.

For $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $\alpha = 1/\sqrt{2}$, $\beta = 1/\sqrt{2}$.

For $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, $\alpha = 1/\sqrt{2}$, $\beta = -1/\sqrt{2}$.

Exam key: If they ask for $H|1\rangle$, beta is negative. If they ask general state, alpha and beta are amplitudes.

Q45. What are Bell states and how do they arise? What is the significance of Bell states and why do we care about them?

Bell states are four maximally entangled two-qubit states:

$$|\Phi^{++}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

$$|\Phi^{--}\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$$

$$|\Psi^{++}\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$$

$$|\Psi^{--}\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$$

They arise by applying H to one qubit and then CNOT between two qubits, with different initial basis states or phase/bit corrections producing the four versions.

Significance: Bell states are the basic entanglement resource for quantum teleportation, quantum key distribution, quantum networking, superdense coding, and tests of non-classical correlations.

Exam key: Bell states are the standard maximally entangled two-qubit states.

Q46. Entanglement is implemented using CNOT, equivalent to XOR in classical logic. Generate a truth table showing the relation between control and target states.

CNOT leaves the control unchanged. The target becomes target XOR control. If control = 0, target stays the same. If control = 1, target flips.

Exam key: CNOT output = (control, target XOR control).

Q47. What is teleportation and how does it work? What is the difference between teleportation and other types of network messaging?

Teleportation transfers an unknown quantum state from Alice to Bob using a shared Bell pair and two classical bits. Alice does not copy or directly send the qubit state. She performs a Bell measurement, sends the two classical outcomes, and Bob applies X/Z corrections to recover the state.

Difference from classical network messaging: classical messages can be copied, amplified, inspected, and resent. An unknown quantum state cannot be copied, and measurement destroys the original state. Teleportation also requires pre-shared entanglement plus a classical channel.

Exam key: Teleportation sends quantum state information without copying the state.

Q48. Assuming Anas and Badr communicate using teleportation across wide distances, what initial setup is needed? Could this setup be done by a third party? What are the requirements?

Initial setup: Anas and Badr must share an entangled Bell pair. Anas holds one qubit of the pair and Badr holds the other. They also need a reliable classical channel so Anas can send the two measurement bits to Badr.

Yes, a third party can prepare and distribute the Bell pair. The requirements are: the distributed qubits must remain coherent, the parties must know which entangled qubits are paired, the channel/distribution must be trusted or verified, and classical communication must be available. In practical long-distance networks, quantum repeaters or entanglement swapping may be needed.

Exam key: Need pre-shared entanglement and classical communication; a third party can distribute entangled pairs.

Q49. What is the significance of teleportation for quantum networking? Why is it so important?

Teleportation is important because it provides a way to move quantum states between distant nodes even though unknown states cannot be copied. It is a core mechanism for quantum repeaters, long-distance quantum communication, distributed quantum computing, and future quantum internet architectures.

Exam key: Teleportation is a building block of the quantum internet.

Q50. Explain using a circuit diagram how Anas can teleport a qubit state to Badr without directly copying the state.

Let q_0 be Anas's unknown qubit, q_1 be Anas's half of the Bell pair, and q_2 be Badr's half.

```

q0 |psi> ----o----H----M c0 -----
      |      |
q1 |0>  --H---X---M c1 --|-----
      |      |
q2 |0>  ---X-----X if c1--Z if c0-- |psi>

```

The original qubit is not copied. Anas measures her qubits, which destroys her original quantum state. Badr reconstructs the state only after receiving the classical bits and applying the corrections.

Exam key: No copying happens; the original is destroyed by measurement.

Q51. What is the task completed in phase one of the teleportation process, and what is the impact of this task on the involved qubits?

In phase one, Anas and Badr create/share the entangled Bell pair. Usually this is done by applying H to one qubit and CNOT from that qubit to the other. The impact is that their two qubits become one entangled resource, meaning their combined state is correlated and cannot be described independently.

If the course divides teleportation after setup, Alice's first active phase is the Bell measurement: she applies CNOT(q_0, q_1), H(q_0), then measures q_0 and q_1 . This collapses Alice's qubits into classical bits and leaves Badr's qubit in one of four related states that can be corrected to $|\psi\rangle$.

Exam key: Phase one may mean Bell-pair setup or Alice Bell measurement; know both descriptions.

Q52. What are the roles of the Z and X gates in the final phase of the teleportation process?

The X and Z gates are Bob's correction gates. X corrects a bit-flip error. Z corrects a phase-flip error. With the common measurement convention:

- If Alice's second measured bit $c_1 = 1$, Bob applies X.
- If Alice's first measured bit $c_0 = 1$, Bob applies Z.

After these corrections, Bob's qubit becomes the original $|\psi\rangle$.

Exam key: X fixes bit flip; Z fixes phase flip.

Q53. What are the results of measurements in the teleportation circuit and how are these transferred to Badr?

Alice measures two qubits and gets one of four classical outcomes: 00, 01, 10, or 11. She sends these two bits to Badr using a classical communication channel such as a normal network link. Badr uses the bits to decide which correction to apply.

Correction table:

00 -> no correction

01 -> apply X

10 -> apply Z

11 -> apply X and Z, order only changes an irrelevant global phase

After correction, Badr's qubit is the original $|\psi\rangle$.

Exam key: The measurement results are classical bits, not quantum data. They are sent over a normal classical channel.

Final Double-Check Checklist

[OK] All 5 quantum tutorial files were covered.

[OK] All explicit questions and table/circuit tasks were answered.

[OK] Hadamard transformations and vector states were corrected and made consistent.

[OK] Entanglement, Bell states, and CNOT truth tables are included.

[OK] Teleportation setup, circuit, measurement outcomes, and X/Z corrections are included.

[OK] Shor, QFT, Grover, and Qiskit custom-gate answers are included.

[OK] The explanations are written in exam-answer style rather than just short definitions.